if $t = \ell + 1$ or $t = T$ and

$$\boldsymbol{d}(t, \ell+3, k, w) + \boldsymbol{d}(\ell+3, t, k, w) = \tilde{\eta_V} \begin{bmatrix} -\boldsymbol{p}_t \mathbb{E}[T^{-1}] \\ \boldsymbol{0} \\ \boldsymbol{0} \end{bmatrix} + \tilde{\eta_V} O_\infty(\mathbb{E}[T^{-1}] N_{\mathrm{trg}} \cdot N^{-1} + N_{\mathrm{trg}} \cdot N^{-2}).$$

otherwise.

Now we are ready to calculate $-\frac{1}{N-N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{\Delta}(k,T)[:, : L+N]$ as

$$\begin{aligned}
&-\frac{1}{N-N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{\Delta}(k,T)[:, 1 : L+N] \\
=&\frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \Bigg\{ -\frac{\tilde{\eta_V}}{N-N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \Bigg[\left(\frac{1}{T} - \frac{1}{T^2}\right) \left(\begin{bmatrix} -\boldsymbol{p}_{\ell+2}\mathbb{E}[T^{-1}] \\ -\boldsymbol{e}_k \mathbb{E}[T^{-1}] \\ -\boldsymbol{e}_w \mathbb{E}[T^{-1}] \end{bmatrix} + \begin{bmatrix} -\boldsymbol{p}_{\ell+3}\mathbb{E}[T^{-1}] \\ \boldsymbol{0} \\ -\boldsymbol{e}_k \mathbb{E}[T^{-1}] \end{bmatrix} \right) \\
&+ \frac{1}{T^2} \begin{bmatrix} \boldsymbol{p}_{\ell+2}\mathbb{E}[T^{-1}] + \boldsymbol{p}_{\ell+3}\mathbb{E}[T^{-1}] \\ \boldsymbol{e}_k \mathbb{E}[T^{-1}] \\ \boldsymbol{e}_k \mathbb{E}[T^{-1}] + \boldsymbol{e}_w \mathbb{E}[T^{-1}] \end{bmatrix} + \frac{2}{T^2} \begin{bmatrix} \boldsymbol{1}_{1:T \setminus \{\ell+2, \ell+3\}} \mathbb{E}[T^{-1}] \\ 2\boldsymbol{e}_w \mathbb{E}[T^{-1}] \\ \boldsymbol{0} \end{bmatrix} \Bigg] \begin{bmatrix} \boldsymbol{p}_T^\top & \boldsymbol{e}_w^\top \end{bmatrix} \Bigg\} \\
&+ \tilde{\eta_V} \mathbb{E}[T^{-1}] O(N_{\mathrm{trg}} \cdot N^{-1}) + \tilde{\eta_V} O(N_{\mathrm{trg}} \cdot N^{-2}) \\
=&\frac{1}{N_{\mathrm{trg}}} \sum_{w=1}^{N_{\mathrm{trg}}} \Bigg\{ \frac{\tilde{\eta_V}}{T} \begin{bmatrix} \boldsymbol{p}_{\ell+2}\mathbb{E}[T^{-1}] + \boldsymbol{p}_{\ell+3}\mathbb{E}[T^{-1}] \\ \boldsymbol{0} \\ \boldsymbol{e}_w \mathbb{E}[T^{-1}] \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_T^\top & \boldsymbol{e}_w^\top \end{bmatrix} \\
&\qquad - \frac{2\tilde{\eta_V}}{T^2} \begin{bmatrix} \boldsymbol{1}_{1:T \setminus \{\ell+2, \ell+3\}} \mathbb{E}[T^{-1}] \\ 2\boldsymbol{e}_w \mathbb{E}[T^{-1}] \\ \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{p}_T^\top & \boldsymbol{e}_w^\top \end{bmatrix} \Bigg\} \\
&\qquad + \tilde{\eta_V} \mathbb{E}[T^{-1}] O_\infty(N_{\mathrm{trg}} \cdot N^{-1}) + \tilde{\eta_V} O_\infty(N_{\mathrm{trg}} \cdot N^{-2}),
\end{aligned}$$

which concludes the proof together with $\mathbb{E}[T^{-1}] \gtrsim N^{-1}$ from Assumption 1. □

We can also bound the last column:

Lemma 12. *It holds that*

$$\begin{aligned}
\boldsymbol{W}_{KQ}^*[:, L+N+1 :] =& -\frac{1}{N-N_{\mathrm{trg}}} \sum_{k=N_{\mathrm{trg}}+1}^{N} \boldsymbol{\Delta}(k,T)[:, L+N+1 :] \\
=& \tilde{\eta_V} \eta_{KQ} \mathbb{E}_{\ell'}[T(\ell')^{-1}] O_\infty(N_{\mathrm{trg}} \cdot N^{-1}).
\end{aligned}$$

Proof. Note that

$$\begin{aligned}
&\boldsymbol{\Delta}(k,T)[:, L+N+1 :] \\
=&\frac{1}{T} \sum_{t=1}^{T} \Big(\mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{e}_{z_{T-1}}^\top \right] - \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) \boldsymbol{e}_{z_{T-1}}^\top | z_{T+1} = k \right] \Big) \\
=&\frac{1}{N-N_{\mathrm{trg}}} \sum_{l=N_{\mathrm{trg}}+1}^{N} \frac{1}{T} \sum_{t=1}^{T} (\mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) | z_{T-1} = l \right] \\
&\qquad - \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) | z_{T-1} = l, z_{T+1} = k \right]) \boldsymbol{e}_l^\top.
\end{aligned}$$

Then the assertion immediately follows from the fact that each entry of

$$\mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) | z_{T-1} = l \right] - \mathbb{E}_{\boldsymbol{X}} \left[\boldsymbol{e}_k^\top \boldsymbol{W}_V^* \boldsymbol{x}_t \cdot (\boldsymbol{x}_t - \bar{\boldsymbol{x}}_{1:T}) | z_{T-1} = l, z_{T+1} = k \right]$$

is upper bounded by $\tilde{\eta_V} \mathbb{E}[T^{-1}]$ up to constant, from Lemma 9. □

B.3 Max-sum Ratio and Algorithm Selection

Now we are ready to establish analysis on transformer's algorithm selection based on max-sum ratio. From Lemmas 11 and 12, it holds that

$$\begin{aligned}
&\boldsymbol{W}_{KQ}^* \\
=&\tilde{\eta}\frac{1}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}\mathbb{E}\left[\left\{\begin{bmatrix}(T^{-1}+2T^{-2})(\boldsymbol{p}_{\ell+2}+\boldsymbol{p}_{\ell+3})\\ \boldsymbol{0}\\ T^{-1}\boldsymbol{e}_w\end{bmatrix}-2T^{-2}\begin{bmatrix}\boldsymbol{1}_{1:T}\\ 2\boldsymbol{e}_w\\ \boldsymbol{0}\end{bmatrix}\right\}\begin{bmatrix}\boldsymbol{p}_T^\top & \boldsymbol{e}_w^\top & \boldsymbol{0}_N^\top\end{bmatrix}\right] \\
&+O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1})
\end{aligned}$$

where $\tilde{\eta}=\eta_V\eta_{KQ}\mathbb{E}[T^{-1}]$.

Assume that a test sequence $z=[z_1,\ldots,z_{T^*},z_{T^*+1}]$ is made from subtext lengths (ℓ_1^*,ℓ_2^*) (hence $T^*=\ell_1^*+\ell_2^*+3$). Now let $q_\lambda=\mathrm{P}[\ell=\lambda]$ and $q^*=\mathrm{P}[2\ell+3=T^*]$ respectively (probability is defined by pretraining distribution). If the trigger $\boldsymbol{x}_{T^*}$ satisfies $\boldsymbol{x}_{T^*}=w^*$, it holds that

$$\begin{aligned}
\boldsymbol{W}_{KQ}^*\boldsymbol{x}_{T^*}=&\tilde{\eta}\frac{q^*}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}\left(\begin{bmatrix}((T^*)^{-1}+2(T^*)^{-2})(\boldsymbol{p}_{\ell^*+2}+\boldsymbol{p}_{\ell^*+3})\\ \boldsymbol{0}\\ (T^*)^{-1}\boldsymbol{e}_w\end{bmatrix}-2(T^*)^{-2}\begin{bmatrix}\boldsymbol{1}_{1:T^*}\\ 2\boldsymbol{e}_w\\ \boldsymbol{0}\end{bmatrix}\right) \\
&+\tilde{\eta}\frac{1}{N_{\mathrm{trg}}}\mathbb{E}\left[\begin{bmatrix}(T^{-1}+2T^{-2})(\boldsymbol{p}_{\ell+2}+\boldsymbol{p}_{\ell+3})\\ \boldsymbol{0}\\ T^{-1}\boldsymbol{e}_{w^*}\end{bmatrix}-2T^{-2}\begin{bmatrix}\boldsymbol{1}_{1:T}\\ 2\boldsymbol{e}_{w^*}\\ \boldsymbol{0}\end{bmatrix}\right]+O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}),
\end{aligned}$$

where $\ell^*=(T^*-3)/2$ — if such ℓ^* is not an integer, then we do not define ℓ^* (in such case $q^*=0$ holds and we don't need to define such a quantity).

Hence, for any t we can calculate the attention logit as

$$\begin{aligned}
s_t\coloneqq&\boldsymbol{x}_t^\top\boldsymbol{W}_{KQ}^*\boldsymbol{x}_{T^*} \\
=&\tilde{\eta}\frac{q^*}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}\Big(((T^*)^{-1}+2(T^*)^{-2})(\mathbb{I}(t=\ell^*+2)+\mathbb{I}(t=\ell^*+3)) \\
&+(T^*)^{-1}(\mathbb{I}(z_{t-1}=w))-2(T^*)^{-2}(\mathbb{I}(t\le T^*)+2\mathbb{I}(z_t=w))\Big) \\
&+\tilde{\eta}\frac{1}{N_{\mathrm{trg}}}\Big((T(t-2)^{-1}+2T(t-2)^{-2})q_{t-2}+(T(t-3)^{-1}+2T(t-3)^{-2})q_{t-3} \\
&+\mathbb{E}[T^{-1}]\mathbb{I}(z_{t-1}=w^*)-2\mathbb{E}[T^{-2}\mathbb{I}(t\le T)]-4\mathbb{E}[T^{-2}]\mathbb{I}(z_t=w^*)\Big) \\
&+O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}).
\end{aligned}\tag{B.3}$$

We begin with showing that if max-sum ratio is not sufficiently large, we can construct an OOD test sequence z^* such that transformer mistakenly use the positional shortcut:

Lemma 13. *If it holds that*

$$\frac{\max_\ell q_\ell\ell^{-1}}{\sum_\ell q_\ell\ell^{-1}}\ge\epsilon(N_{\mathrm{trg}})$$

where $\epsilon(N_{\mathrm{trg}})=\Theta(N_{\mathrm{trg}}^{-1})$, there exists an OOD test sequence such that the pretrained transformer via Algorithm 1 fails to generalize.

Proof. Assume that

$$z^*=[\underbrace{u,u,\ldots,u}_{\ell_1^*},w^*,v,\underbrace{u,u,\ldots,u}_{\ell_2^*},w^*,v].$$

and $T^*=\ell_1^*+\ell_2^*+3=2\ell^*+3$ where $\ell^*=\arg\max_\ell q(\ell)\ell^{-1}$. Furthermore, we assume

$$\ell_1^* \notin \{\ell^*-1, \ell^*, \ell^*+1, \ell^*+2\}. \tag{B.4}$$

Since we have Assumption 2, there exists $\ell_1^* \geq 1$ such that (B.4) holds. We show the following sub-lemma:

Lemma 14. *There exists $\epsilon_1(N_{\mathrm{trg}}) = \Theta(N_{\mathrm{trg}}^{-1})$ such that if*

$$\frac{\max_\ell q_\ell \ell^{-1}}{\sum_\ell q_\ell \ell^{-1}} \geq \epsilon_1(N_{\mathrm{trg}}),$$

then

$$s_{\ell_1^*+1}, s_{\ell_1^*+2}, s_{\ell_1^*+3}, s_{T^*} \leq \frac{1}{2} s_{\ell^*+2}.$$

Proof. Here we show $2s_{\ell_1^*+2} \leq s_{\ell^*+2}$ — other properties can be deduced in the same vain.

Note that, from (B.3),

$$\begin{aligned}
s_{\ell_1^*+2} &\leq \tilde{\eta}\frac{q^*}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}[(T^*)^{-1}\mathbb{I}(w=w^*)] \\
&+ \frac{\tilde{\eta}}{N_{\mathrm{trg}}}[((2\ell_1^*+3)^{-1}+2(2\ell_1^*+3)^{-2})q_{\ell_1^*} + ((2\ell_1^*+1)^{-1}+2(2\ell_1^*+1)^{-2})q_{\ell_1^*-1} + \mathbb{E}[T^{-1}]] \\
&+ O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}) \\
&\leq \frac{\tilde{\eta}q^*}{N_{\mathrm{trg}}}(T^*)^{-1} + \frac{6\tilde{\eta}q^*}{N_{\mathrm{trg}}}(T^*)^{-1} + \frac{\tilde{\eta}}{N_{\mathrm{trg}}}\mathbb{E}[T^{-1}] + O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}).
\end{aligned}$$

On the other hand, it holds that

$$\begin{aligned}
s_{\ell^*+2} &\geq \tilde{\eta}\frac{q^*}{N_{\mathrm{trg}}}\sum_{w=1}^{N_{\mathrm{trg}}}[(T^*)^{-1}+2(T^*)^{-2}-2(T^*)^{-2}(\mathbb{I}(\ell^*+2\leq T^*)+2\mathbb{I}(z_{\ell^*+2}=w))] \\
&+ \frac{\tilde{\eta}}{N_{\mathrm{trg}}}[((2\ell^*+3)^{-1}+2(2\ell^*+3)^{-2})q_{\ell^*} + ((2\ell^*+1)^{-1}+2(2\ell^*+1)^{-2})q_{\ell^*-1} - 6\mathbb{E}[T^{-2}]] \\
&+ O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}) \\
&\geq \tilde{\eta}q^*(T^*)^{-1} - 6\tilde{\eta}q^*(T^*)^{-2} - 6\frac{\tilde{\eta}}{N_{\mathrm{trg}}}\mathbb{E}[T^{-2}] + O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}).
\end{aligned}$$

Note that $T^{-2} \leq \frac{1}{10}T^{-1}$ holds from Assumption 2. Therefore,

$$\frac{1}{2}s_{\ell^*+2} - s_{\ell_1^*+2} \geq \frac{1}{5}\tilde{\eta}q^*(T^*)^{-1} - \frac{13}{10}\frac{\tilde{\eta}}{N_{\mathrm{trg}}}\mathbb{E}[T^{-1}] - \frac{7\tilde{\eta}q^*}{N_{\mathrm{trg}}}(T^*)^{-1} + O_\infty(\tilde{\eta}N_{\mathrm{trg}}\cdot N^{-1}).$$

Together with Assumption 1, if the max-sum ratio is $\Omega(N_{\mathrm{trg}}^{-1})$, we obtain $2s_{\ell_1^*+2} \leq s_{\ell^*+2}$ as desired. □

Since now we have Lemma 14, when

$$\tilde{\eta}_V\eta_{KQ} \gtrsim C\log N\frac{N^2}{N_{\mathrm{trg}}^3} \gtrsim C\log N\cdot\frac{N_{\mathrm{trg}}}{\mathbb{E}[T^{-1}]^2} \tag{B.5}$$

for a sufficiently large C we obtain $\exp s_t/\exp s_{\ell^*+2} \leq \exp\{-C\log N\} = N^{-C}$ where $t = \ell_1^*+1, \ell_1^*+2, \ell_1^*+3$ and T^*. This immediately implies that $\boldsymbol{X}_{1:T^*}\mathrm{Softmax}(\boldsymbol{X}_{1:T^*}^\top\boldsymbol{W}_{KQ}\boldsymbol{x}_{T^*}) = \sum_{k\neq\ell^*+1,\ell^*+2,\ell^*+3,T^*}\alpha_k\boldsymbol{x}_k + O_\infty(N^{-C'})$ for a sufficiently large C' where $\sum_{k\neq\ell^*+1,\ell^*+2,\ell^*+3,T^*}\alpha_k \geq 1-N^{-C'}$. Therefore, we get

$$\boldsymbol{X}_{1:T^*}\mathrm{Softmax}(\boldsymbol{X}_{1:T^*}^\top\boldsymbol{W}_{KQ}\boldsymbol{x}_{T^*}) = (1-N^{-C'})\begin{bmatrix}*\\ \boldsymbol{e}_u\\ \boldsymbol{e}_u\end{bmatrix} + N^{-C'}\begin{bmatrix}*\\ *\\ *\end{bmatrix}.$$